

Understanding the Derivative

1.1 How do we measure velocity?

Motivating Questions

- How is the average velocity of a moving object connected to the values of its position function?
 - How do we interpret the average velocity of an object geometrically on the graph of its position function?
 - How is the notion of instantaneous velocity connected to average velocity?
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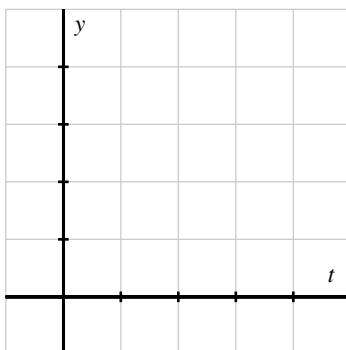
1.1.1 Introduction

Calculus can be viewed broadly as the study of change. A natural and important question to ask about any changing quantity is “how fast is the quantity changing?”

We begin with a simple problem: a ball is tossed straight up in the air. How is the ball moving? Questions like this one are central to our study of differential calculus.

Preview Activity 1.1.1. Suppose that the height s of a ball at time t (in seconds) is given in feet by the formula $s(t) = 64 - 16(t - 1)^2$.

- (a) Construct a graph of $y = s(t)$ on the time interval $0 \leq t \leq 3$. Label at least six distinct points on the graph, including the three points showing when the ball was released, when the ball reaches its highest point, and when the ball lands.



- (b) Describe the behavior of the ball on the time interval $0 < t < 1$ and on time interval $1 < t < 3$. What occurs at the instant $t = 1$?
- (c) Consider the expression

$$AV_{[0.5,1]} = \frac{s(1) - s(0.5)}{1 - 0.5}.$$

Compute the value of $AV_{[0.5,1]}$. What does this value measure on the graph? What does this value tell us about the motion of the ball? In particular, what are the units on $AV_{[0.5,1]}$?

1.1.2 Position and average velocity

Any moving object has a *position* that can be considered a function of *time*. When the motion is along a straight line, the position is given by a single variable, which we denote by $s(t)$. For example, $s(t)$ might give the mile marker of a car traveling on a straight highway at time t in hours. Similarly, the function s described in Preview Activity 1.1.1 is a position function, where position is measured vertically relative to the ground.

On any time interval, a moving object also has an *average velocity*. For example, to compute a car's average velocity we divide the number of miles traveled by the time elapsed, which gives the velocity in *miles per hour*. Similarly, the value of $AV_{[0.5,1]}$ in Preview Activity 1.1.1 gave the average velocity of the ball on the time interval $[0.5, 1]$, measured in feet per second.

In general, we make the following definition:

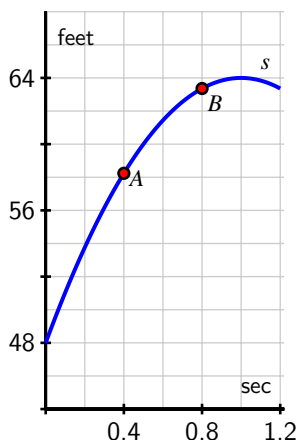
Average Velocity.

For an object moving in a straight line with position function $s(t)$, the *average velocity* of the object on the interval from $t = a$ to $t = b$, denoted $AV_{[a,b]}$, is given by the formula

$$AV_{[a,b]} = \frac{s(b) - s(a)}{b - a}.$$

Note well: the units on $AV_{[a,b]}$ are “units of s per unit of t ,” such as “miles per hour” or “feet per second.”

Activity 1.1.2. The following questions concern the position function given by $s(t) = 64 - 16(t - 1)^2$, considered in Preview Activity 1.1.1, and plotted in the given figure.



- Compute the average velocity of the ball on each of the following time intervals: $[0.4, 0.8]$, $[0.7, 0.8]$, $[0.79, 0.8]$, $[0.799, 0.8]$, $[0.8, 1.2]$, $[0.8, 0.9]$, $[0.8, 0.81]$, $[0.8, 0.801]$. Include units for each value.
- On the graph provided in (a), sketch the line that passes through the points $A = (0.4, s(0.4))$ and $B = (0.8, s(0.8))$. What is the meaning of the slope of this line? In light of this meaning, what is a geometric way to interpret each of the values computed in the preceding question?
- Use a graphing utility to plot the graph of $s(t) = 64 - 16(t - 1)^2$ on an interval containing the value $t = 0.8$. Then, zoom in repeatedly on the point $(0.8, s(0.8))$. What do you observe about how the graph appears as you view it more and more closely?
- What do you conjecture is the velocity of the ball at the instant $t = 0.8$? Why?

1.1.3 Instantaneous Velocity

Whether we are driving a car, riding a bike, or throwing a ball, we have an intuitive sense that a moving object has a velocity at any given moment — a number that measures how fast the object is moving *right now*. For instance, a car’s speedometer tells the driver the car’s velocity at any given instant. In fact, the velocity on a speedometer is really an average velocity that is computed over a very small time interval. If we let the time interval over which average velocity is computed become shorter and shorter, we can progress from average velocity to *instantaneous* velocity.

Informally, we define the *instantaneous velocity* of a moving object at time $t = a$ to be the value that the average velocity approaches as we take smaller and smaller intervals of time containing $t = a$. We will develop a more formal definition of instantaneous velocity soon, and this definition will be the foundation of much of our work in calculus. For now, it is fine to think of instantaneous velocity as follows: take average velocities on smaller and smaller time intervals around a specific point. If those average velocities approach a single number, then that number will be the instantaneous velocity at that point.

Activity 1.1.3. Each of the following questions concern $s(t) = 64 - 16(t - 1)^2$, the position function from Preview Activity 1.1.1.

- Compute the average velocity of the ball on the time interval $[1.5, 2]$. What is different between this value and the average velocity on the interval $[0, 0.5]$?
- Use appropriate computing technology to estimate the instantaneous velocity of the ball at $t = 1.5$. Likewise, estimate the instantaneous velocity of the ball at $t = 2$. Which value is greater?
- How is the sign of the instantaneous velocity of the ball related to its behavior at a given point in time? That is, what does positive instantaneous velocity tell you the ball is doing? Negative instantaneous velocity?
- Without doing any computations, what do you expect to be the instantaneous velocity of the ball at $t = 1$? Why?

At this point we have started to see a close connection between average velocity and instantaneous velocity. Each is connected not only to the physical behavior of the moving object but also to the geometric behavior of the graph of the position function. We are interested in computing average velocities on the interval $[a, b]$ for smaller and smaller intervals. In order to make the link between average and instantaneous velocity more formal, think of the value b as $b = a + h$, where h is a small (non-zero) number that is allowed to vary. Then the average velocity of the object on the interval $[a, a + h]$ is

$$AV_{[a, a+h]} = \frac{s(a+h) - s(a)}{h},$$

with the denominator being simply h because $(a+h) - a = h$. Note that when $h < 0$, $AV_{[a, a+h]}$ measures the average velocity on the interval $[a+h, a]$.

To find the instantaneous velocity at $t = a$, we investigate what happens as the value of h approaches zero.

Example 1.1.1 Computing instantaneous velocity for a falling ball. The position function for a falling ball is given by $s(t) = 16 - 16t^2$ (where s is measured in feet and t in seconds).

- Find an expression for the average velocity of the ball on a time interval of the form $[0.5, 0.5 + h]$ where $-0.5 < h < 0.5$ and $h \neq 0$.
- Use this expression to compute the average velocity on $[0.5, 0.75]$ and $[0.4, 0.5]$.
- Make a conjecture about the instantaneous velocity at $t = 0.5$.